

Web Manual-01

Open terminal in Ubuntu and run:

commands

```
sudo apt update && sudo apt upgrade -y
```

CHECKPOINT-1 (2 Marks) — Install Apache + Default Page

1. Install Apache

```
sudo apt install apache2 -y
```

2. Allow Apache in firewall

```
sudo ufw allow 'Apache'
```

```
sudo ufw reload
```

3. Check status

```
sudo systemctl status apache2
```

4. Map domain to localhost

```
sudo nano /etc/hosts
```

```
127.0.0.1  webserverlab.com
```

CHECKPOINT-2 (5 Marks) — Single Virtual Host (example.com)

1. Create directory

```
sudo mkdir -p /var/www/example.com/html
```

2. Set ownership

```
sudo chown -R $USER:$USER /var/www/example.com/html
```

3. Set permissions

```
sudo chmod -R 755 /var/www/example.com
```

4. Create index.html

```
nano /var/www/example.com/html/index.html
```

```
<html>
```

```
<head>
```

```
<title>Welcome to Example.com!</title>
```

```
</head>
```

```
<body>
```

```
<h1>Success! The example.com server block is working!</h1>
```

```
</body>
```

```
</html>
```

5. Create virtual host file

```
sudo nano /etc/apache2/sites-available/example.com.conf
```

```
<VirtualHost *:80>
```

```
ServerAdmin admin@example.com
```

```
ServerName example.com
```

```
ServerAlias www.example.com
```

```
DocumentRoot /var/www/example.com/html
```

```
ErrorLog ${APACHE_LOG_DIR}/error.log
```

```
CustomLog ${APACHE_LOG_DIR}/access.log combined
```

```
</VirtualHost>
```

6. Enable site and disable default

```
sudo a2ensite example.com.conf
```

```
sudo a2dissite 000-default.conf
```

7. Test config

```
sudo apache2ctl configtest
```

8. Restart Apache

```
sudo systemctl restart apache2
```

9. Add domain to /etc/hosts

```
sudo nano /etc/hosts
```

CHECKPOINT-3 (4 Marks) — Explain What Happens

```
sudo a2ensite 000-default.conf
```

```
sudo systemctl restart apache2
```

CHECKPOINT-4 (4 Marks) — Multiple Virtual Hosts

```
sudo mkdir -p /var/www/anotherhost.com/html
```

```
sudo chown -R $USER:$USER /var/www/anotherhost.com/html
```

```
sudo chmod -R 755 /var/www/anotherhost.com
```

```
nano /var/www/anothervhost.com/html/index.html
```

```
<html>

  <head><title>Another Site</title></head>

  <body>

    <h1 style="color: red;">Welcome to AnotherVHost.com!</h1>

    <p>This is a different virtual host.</p>

  </body>

</html>
```

```
sudo nano /etc/apache2/sites-available/anothervhost.com.conf
```

```
<VirtualHost *:80>

  ServerAdmin admin@anothervhost.com

  ServerName anothervhost.com

  ServerAlias www.anothervhost.com

  DocumentRoot /var/www/anothervhost.com/html

  ErrorLog ${APACHE_LOG_DIR}/error.log

  CustomLog ${APACHE_LOG_DIR}/access.log combined

</VirtualHost>
```

```
sudo a2ensite anothervhost.com.conf
```

```
sudo apache2ctl configtest
```

```
sudo systemctl restart apache2
```

```
sudo nano /etc/hosts
```

```
127.0.0.1  anothervhost.com
```

```
127.0.0.1  www.anothervhost.com
```

CHECKPOINT-5 (5 Marks) — Two Dynamic Websites with HTML + JavaScript Forms

Site 1: Calculator (example.com)

```
nano /var/www/example.com/html/index.html
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Calculator - example.com</title>
```

```
</head>
```

```
<body>
```

```
    <h1>JavaScript Calculator</h1>
```

```
    <form name="calc">
```

```
        <input type="text" name="num1" placeholder="Number 1">
```

```
        <input type="text" name="num2" placeholder="Number 2">
```

```
        <br><br>
```

```
        <button type="button" onclick="add()">Add</button>
```

```
        <button type="button" onclick="sub()">Subtract</button>
```

```
        <button type="button" onclick="mul()">Multiply</button>
```

```
<button type="button" onclick="div()">Divide</button>
```

```
<br><br>
```

```
<h3>Result: <span id="result">0</span></h3>
```

```
</form>
```

```
<script>
```

```
function add() {
```

```
    let a = parseFloat(document.calc.num1.value);
```

```
    let b = parseFloat(document.calc.num2.value);
```

```
    document.getElementById("result").innerHTML = a + b;
```

```
}
```

```
function sub() {
```

```
    let a = parseFloat(document.calc.num1.value);
```

```
    let b = parseFloat(document.calc.num2.value);
```

```
    document.getElementById("result").innerHTML = a - b;
```

```
}
```

```
function mul() {
```

```
    let a = parseFloat(document.calc.num1.value);
```

```
    let b = parseFloat(document.calc.num2.value);
```

```
    document.getElementById("result").innerHTML = a * b;
```

```
}
```

```
function div() {
```

```
    let a = parseFloat(document.calc.num1.value);
```

```
        let b = parseFloat(document.calc.num2.value);

        document.getElementById("result").innerHTML = (b !== 0) ? (a / b) : "Error";

    }

</script>

</body>

</html>
```

Site 2: Grade Calculator (anothervhost.com)

```
<!DOCTYPE html>

<html>

<head>

    <title>Grade Calculator</title>

    <style>body { font-family: Arial; background: #f0f8ff;}</style>

</head>

<body>

    <h1>Student Grade Calculator</h1>

    <form name="grade">

        Marks 1: <input type="number" name="m1"><br><br>

        Marks 2: <input type="number" name="m2"><br><br>

        Marks 3: <input type="number" name="m3"><br><br>

        <button type="button" onclick="calculate()">Calculate Grade</button>

    </form>

    <h2>Average: <span id="avg">0</span></h2>

    <h2>Grade: <span id="grade">F</span></h2>
```

```
<script>
```

```
function calculate() {
```

```
    let a = parseFloat(document.grade.m1.value);
```

```
    let b = parseFloat(document.grade.m2.value);
```

```
    let c = parseFloat(document.grade.m3.value);
```

```
    let avg = (a + b + c) / 3;
```

```
    document.getElementById("avg").innerHTML = avg.toFixed(2);
```

```
    let g = "F";
```

```
    if (avg >= 80) g = "A+";
```

```
    else if (avg >= 70) g = "A";
```

```
    else if (avg >= 60) g = "B";
```

```
    else if (avg >= 50) g = "C";
```

```
    else if (avg >= 40) g = "D";
```

```
    document.getElementById("grade").innerHTML = g;
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

```
sudo systemctl restart apache2
```


